

LECTURER

- Duc Ta --- dta@sfsu.edu

Office Location:

Thornton Hall 432

Office Hours:

MoWe between classes and by appointment

Communication: https://csc215.ducta.net/02-QUESTIONS_WhoToContact.pdf**COURSE**

- CSC 215
- Section PING
- File Manager (sfsu/student):*
- Forums (on Canvas):*

Intermediate Computer Programming, **3-unit Lecture** and **1-unit Seminar**

MoWe 8:00 AM - 09:15 AM, Thornton Hall 432

<https://csc215.ducta.net><https://canvas.sfsu.edu>**BULLETIN DESCRIPTION**

Design, implementation, testing, debugging, maintenance, and documentation of Java programs. Algorithms, programming concepts, and data types in Java. Concepts of object-oriented programming; numerical and non-numerical problems. Required for Computer Science majors and minors.

COURSE PREREQUISITES

- ✗ CSC 101 with a grade of C or better.

COURSE OBJECTIVES

At the end of this course, students will

- ✓ Demonstrate basic programming skills in Java.
- ✓ Demonstrate understanding of concepts of algorithmic problem solving and design of computer programs.
- ✓ Demonstrate the ability to write, debug, and execute small and medium-sized Java programs.
- ✓ Demonstrate their readiness to continue the study of object-oriented programming and computer algorithms.

COURSE REQUIREMENTS

A student will be dropped from the course if:

- The student failed to show up and participate fully in the **first** and **second** class meetings.
- The student **does not have** the course prerequisites.
- The student's grade is **incomplete** on either assignment DL-Approval-01 or assignment DL-Approval-02.
- The student's grade is **0** (Zero/no effort) on either the assignment Canvas or assignment 01.

GRADES

Grades will be awarded based on the following breakdown:

- Discussion Section: 125 pts** **In-person** attendance in the Discussion Section (Seminar) is **MANDATORY**.
- Advising Assignment: 25 pts
- Other Assignments: 275 pts
- Midterm Exams, 1 exam: 25 pts
- Final Exam **must-take**, or course grade F: 50 pts
- Extended deadlines & extra credit, if any, must be offered publicly and fairly to **ALL** (not one or a few) students.

Grades will be determined on the basis of the total possible points of 500 and extra credits received. Letter grade scale:

- A: 90-100% B: 80-89% C: 70-79% D: 60-69% F: Below 60%
- Or A: 450-500 points B: 400-449 points C: 350-399 points D: 300-349 points F: Below 300 points
- WU: SFSU-defined
- A and a Letter of Recommendation: 100+% (500+ points) and make the course a better experience for everyone.

ATTENDANCE

- Students are responsible for all material covered in class. It may be a good idea to have a Study Buddy.
- Course announcements and forums are on Canvas. Course materials are organized on the course's File Manager.

LEARNING ACTIVITIES

- Detailed descriptions and instructions are to be given in class for each learning activity.
- To make learning activities meaningful and useful, please engage. Different points of view are very welcome.

STUDENT RESOURCES

- Recommended Textbook: Y. Daniel Liang, Introduction to Java Programming, Comprehensive, 12th edition, Pearson.
- Recommended IDEs: JetBrains IntelliJ IDEA (default course IDE) and your choices. The latest versions.

- Recommended Services, *please visit the File Manager for more recommended services:*
 - CS Department Office, (415) 338-2156 <https://cs.sfsu.edu/>
 - SFSU, ITS, Software and Hardware <https://its.sfsu.edu/services>
 - SFSU, Library, Extended Laptops Checkout <https://library.sfsu.edu/extended-laptops-checkout>

TENTATIVE SCHEDULE

In-person attendance in the Discussion Section is **MANDATORY**. The Discussion Section is worth 1 unit out of the 4 total CSC 215 units. The Lecture is worth 3 units. The CSC 215 Discussion Leaders opened their Discussion sessions and openly invited more CSC 215 students to visit their meetings. **13 x OPEN Invitations** are available weekly. Please visit **DL.ducta.net** regularly. For more volunteer tutoring, please see **tutoring.ducta.net** regularly.

DATE	TOPIC	READING and ASSIGNMENT
WEEK 00 <i>Before Jan 26</i>	- Pre-Semester Preparation <i>Optional pre-semester meetings, orientations, email communication, and self-preparation.</i>	Explore File Manager and Canvas ASMT Canvas Out, ASMT 01 Out ASMT DL-Approval-01 Out ASMT DL-Approval-02 Out
WEEK 01+02+03 <i>Jan 26+28 Feb 02+04+09+11</i>	- Welcome and Introduction <i>Introduction, CSC 215 How-tos, CSC 215 Community How-tos, and CSC 101 Review</i> <i>Switch statement and switch expression, lambda expression, operation counting...</i>	CHAPTER 01-02-03-04-05-06 ASMT Canvas Due, ASMT 01 Due ASMT DL-Approval-01 Due ASMT 02 Out
WEEK 04 <i>Feb 16 + 18</i>	- Single-Dimensional Arrays and Multidimensional Arrays <i>Array, element, index, declare/create/initialize, display, sum, find, max, random, shift, simplify, copy, search, sort...</i>	CHAPTER 07 Independent Reading ASMT DL-Approval-02 Due
WEEK 05 <i>Feb 23 + 25</i>	- Multidimensional Arrays and Irregular Ragged/Jagged Arrays <i>Array, element, index, declare/create/initialize, display, sum, find, max, random, shift, simplify, copy, search, sort...</i>	CHAPTER 08 Independent Reading ASMT 02 Due, ASMT 03 Out
WEEK 06 <i>Mar 02 + 04</i>	- Objects & Classes and Object-Oriented Thinking 1 <i>Object, class, UML, constructor, construct, access, modification, reference...</i>	CHAPTER 09+10 ASMT Advising Out
WEEK 07 <i>Mar 09 + 11</i>	- Objects & Classes and Object-Oriented Thinking 2 <i>Instances, static, getter/accessor, setter/mutator, encapsulation, private, public, paradigms...</i> - Midterm Exam Review	CHAPTER 09+10 Independent Reading ASMT Advising Due ASMT 03 Due, ASMT 04 Out
WEEK 08 <i>Mar 16 + 18</i>	- Midterm Exam Review and - MIDTERM Exam – Mar 18	Independent Reading ASMT 04 Due, ASMT 05 Out
WEEK 09	Spring '26 Recess; No Classes; Campus Open	No Classes
WEEK 10 <i>Mar 30 + Apr 01</i>	- Inheritance and Polymorphism 1 <i>Subclass, superclass, Inheritance IS-A, Composition HAS-A, this, super, overwrite, overload, polymorphism...</i>	CHAPTER 11 Independent Reading
WEEK 11 <i>Apr 06 + 08</i>	- Inheritance and Polymorphism 2 <i>Generics, early/static/compile-time binding, late/dynamic/run-time binding, object type down-casting and up-casting, superclass-subclass access to objects and methods and fields, access control of subclasses within and outside packages, equals and compareTo...</i>	CHAPTER 11 Independent Reading ASMT 05 Due, ASMT 06 Out
WEEK 12 <i>Apr 13 + 15</i>	- Exception Handling and Text I/O <i>Exceptions, debugging, file input, and output...</i>	CHAPTER 12 Independent Reading
WEEK 13 <i>Apr 20 + 22</i>	- Abstract and Interface <i>Abstract, protected, single implementation, multiple implementations...</i>	CHAPTER 13 ASMT 06 Due, ASMT 07 Out
WEEK 14 <i>Apr 27 + 29</i>	- Abstract and Interface <i>Abstract, protected, single implementation, multiple implementations...</i>	CHAPTER 13 Independent Reading
WEEK 15 <i>May 04 + 06</i>	- Sealed Classes and Advanced OOP Topics <i>Sealed, non-sealed, permits, final, nested class, collision, shadowing...</i>	ASMT 07 Due Independent Reading
WEEK 16 <i>May 11 + 13</i>	- Advanced Topics, CSC 220 Preparation, and Final Exam Review <i>Switch expression, ArrayList, StringBuilder, ANSI, Date and Time, internationalization, data structures, sorting, recursion...</i>	Independent Reading
WEEK 17	FINAL Exam: Monday, May 18, 08:00 AM - 10:00 AM	

*** The course lecturer reserves the right to change any of these policies at any time during the semester. Any changes will be announced in class and posted in the course forums on Canvas.

STUDENTS WITH DISABILITIES

Any students with disabilities or other special needs who need special accommodations in this course are invited to share these concerns or requests with the instructor. The Disability Programs and Resource Center (DPRC) is available to facilitate the reasonable accommodations process. The DPRC is located in the Student Service Building and can be reached by telephone (voice/TTY 415-338-2472) or by email (dprc@sfsu.edu).

STUDENT CONDUCT AND ACADEMIC HONESTY**Course Policy on Student Conduct and Academic Honesty**

All students of this course must agree with and strictly follow:

1. The San Francisco State University's **Code of Student Conduct**: <https://conduct.sfsu.edu/standards>
2. The Computer Science Department's **Student Policies**: <https://cs.sfsu.edu/student-policies>
3. **The Honor Code** of this course is detailed below (in this same document).
4. Caring and constructive behavior which helps build a safe and healthy atmosphere in the class for everyone will be rewarded. Please ask the instructor for more details.
5. An F grade or a "zero" grade will be assigned for an assignment in which a student violates a rule in **the Honor Code**.
6. An F grade will be assigned for the course to a student who cheats on an exam.
7. An F grade will be assigned for the course to a student who poisons the learning or the testing environment of others, who prejudices the effort of others, or who infringes on the rights of others.

The Honor Code

This Honor Code was adopted from the Computer Science Department of Stanford University, <https://cs.stanford.edu/degrees/ug/HonorCode.shtml>, and was updated.

A. The Honor Code is an undertaking of the students, individually and collectively:

- that they will not give or receive aid in examinations; that they will not give or receive unpermitted aid in classwork, in the preparation of reports, or in any other work that is to be used by the instructor as the basis of grading;
- that they will do their share and take an active part in seeing to it that others, as well as themselves, uphold the spirit and letter of the Honor Code.

B. Under the Honor Code, you are obligated to follow all of the following rules:

Rule 1: You must not look at solutions or program code that are not your own.

It is an act of plagiarism to submit work that is copied or derived from the work of others and submitted as your own. For example, using a solution from the Internet or a solution from another student (past or present) or some other source, in part or in whole, that is not your own work is a violation of the Honor Code. Many Honor Code infractions we see make use of solution code found online. The best way to steer clear of this possibility is not to search for online solutions to the programming assignments. Moreover, looking at someone else's solution code in order to determine how to solve the problem yourself is also an infraction of the Honor Code. In essence, you should not be looking at someone else's code in order to solve the problems in this class. This is not an appropriate way to "check your work," "get a hint," or "see alternative approaches."

Rule 2: You must not share your solution code with other students.

In particular, you should not ask anyone to give you a copy of their code or, conversely, give your code to another student who asks you for it. Similarly, you should not discuss your algorithmic strategies to such an extent that you and your collaborators end up turning in the same code. Moreover, you are expected to take reasonable measures to maintain the privacy of your solutions. For example, you should not leave copies of your work on public computers nor post your solution code on a public website.

Rule 3: You must indicate on your submission any assistance you received.

If you received aid while producing your solution, you should indicate from whom you got help (if that person is not the grader or the instructor of this class) and what help you received. A proper citation should specifically identify the source (e.g., person's name, book title, website URL, etc.) and a clear indication of how this assistance influenced your work (be as specific as possible). For example, you might write "I discussed the approach used for sorting numbers in the `sortNumbers` method with Mary Smith." If you make use of such assistance without giving proper credit, you may be guilty of plagiarism.

It is also important to make sure that the assistance you receive consists of general advice that does not cross the boundary into having someone else write the actual code or show you their code. It is fine to discuss ideas and strategies, but you should be careful to write your programs on your own, as indicated in Rules 1 and 2.

C. Please be aware: all submissions are subject to automated plagiarism detection.

The course instructor has no desire to create a climate in which students feel as if they are under suspicion. The entire point of the Honor Code is that we all benefit from working in an atmosphere of mutual trust. Students who deliberately take advantage of that trust, however, poison that atmosphere for everyone.

The course grader employs powerful automated plagiarism detection tools that compare assignment submissions with other submissions from the current and previous semesters. The tools also compare submissions against a wide variety of online solutions. These tools are effective at detecting unusual resemblances in programs, which are then further examined by the grader again. The grader then makes the determination whether submissions are deemed to be potential infractions of the Honor Code and should be referred to the Computer Science department.

D. A final note on collaboration.

In computer science courses, it is usually appropriate to ask others--the grader or the instructor, or other students--for hints and debugging help or to talk generally about problem-solving strategies and program structure. In fact, the course instructor strongly encourages you to seek such assistance when you need it. Discuss ideas together, and respect the rules together, but do the coding on your own.

STUDENT DISCLOSURES OF SEXUAL VIOLENCE

SF State fosters a campus free of sexual violence including sexual harassment, domestic violence, dating violence, stalking, and/or any form of sex or gender discrimination. If you disclose a personal experience as an SF State student, the course instructor is required to notify the [Dean of Students]. To disclose any such violence confidentially, contact:

- The SAFE Place - (415) 338-2208; http://www.sfsu.edu/~safe_plc/
- Counseling and Psychological Services Center - (415) 338-2208; <http://psyservs.sfsu.edu/>
- For more information on your rights and available resources: <http://titleix.sfsu.edu>

COURSE SYLLABUS POLICY, #S23-257

- *Senate Policy*
- *Last review: 03/2023 | Next review: 03/2027*

Check List and Additional Items:

- I. Guidelines covering syllabus use in courses
 - ✓ Guidelines adopted
- II. Basic information for all course syllabi
 - A. Provided
 - B. Provided
 - C. Provided
 - D. Provided
 - E. Provided
 - F. Provided
 - G. Provided
 - H. Provided
 - I. Provided
 - J. Provided
 - K. Provided
 - L. Provided
 - M. Provided

N. Students may not capture audio, photos, or video from class sessions on their own devices without the explicit permission of the instructor and everyone present, unless part of a DPRC-authorized accommodation.

O. Students may not post any course materials to any third-party sites (such as Chegg) or post any recordings, screenshots, audio or chat transcripts in any setting outside the class, and that violations of this are subject to student disciplinary action.

P. Provided
- III. Additional information for syllabi for courses using asynchronous and synchronous delivery
 - ✓ Not applied
- IV. Information for course that meet University-wide requirement (i.e., General Education, SF State Studies, and American Institutions requirements)
 - ✓ Not applied
- V. Statements required by Academic Senate Policy
 - A. Disability access

Students with disabilities who need reasonable accommodations are encouraged to contact the instructor. The [Disability Programs and Resource Center (DPRC)] is available to facilitate the reasonable accommodations process. The [DPRC] is located in the [Student Service Building] and can be reached by telephone [(voice/415-338-2472, video phone/415-335-7210) or by email (dprc@sfsu.edu).]
 - B. Student disclosures of sexual violence

SF State fosters a campus free of sexual violence including sexual harassment, domestic violence, dating violence, stalking, and/or any form of sex or gender discrimination. If you disclose a personal experience as an SF State student, the course instructor is required to notify the [Title IX Coordinator by completing the report form available at <http://titleix.sfsu.edu>, emailing vpsaem@sfsu.edu or calling 338- 2032]. To disclose any such violence confidentially, contact:

[The SAFE Place - (415) 338-2208; http://www.sfsu.edu/~safe_plc/]

[Counseling and Psychological Services Center - (415) 338- 2208; <http://psyserve.sfsu.edu/>]

For more information on your rights and available resources: [<http://titleix.sfsu.edu>]

C. Academic integrity

Academic integrity, the ethical presentation of one's own work in accordance with the rules established for this class, is required. Instances of academic misconduct will be reported to the College in which the course is housed, the Division of Graduate Studies (if a graduate student), and the Office of Student Conduct with the report being kept in those offices until a student earns his/her degree. Any instances of cheating, deceit, fabrication, forgery, plagiarism, unauthorized altering of records or submitting false documents, unauthorized collaboration, unauthorized submission of work previously given credit, or other forms of academic misconduct will be assigned a grade penalty, likely an F or a grade of zero. Failing one or more assignments or examinations for reasons of academic integrity violations may result in a final class grade of F. Students may not withdraw from classes in which they have committed academic misconduct. Consequences for violations of academic integrity may exceed an F on the assignment, examination, or class as determined by the Academic Integrity Review Committee.

Members of our academic community have a responsibility to develop an awareness of academic integrity, to cultivate skills to realize honesty in academic and community work, and to sustain actively academic honor as a core value of our community. Students are expected to engage in behaviors that reflect well upon the university. In addition to attending to one's own actions, the Standards for Student Conduct require that students who witness academic dishonesty notify their faculty/instructor, department chair, or the Office of Student Conduct. Supporting academic integrity enhances the reputation of the University and the value attributed to degrees awarded by the University.

GUIDE FOR S23-257 COURSE SYLLABUS POLICY

- *Senate Policy*

- *Version 2: June 6, 2023*

3-unit lecture in person: This class meets 150 minutes per week in person. In addition to class meetings, students can expect to spend, on average, **AT LEAST** an additional 300 minutes (~ 5 hours) per week on preparation, practice, and assignments, including readings, studying for lectures, coding, practice problem-solving, doing assignments, studying for exams, learning more how to problem solve and how to think like a Computer Scientist, getting job-ready...

1-unit discussion course: Students can expect to spend 150 minutes per week doing work for this class including participating in group meetings, meeting with the discussion leader, and writing progress reports. The weekly schedule will be worked out among the department, the discussion leader, and the student when the semester begins.